

Lab 9: Inheritance & Genetics — Problem Solving

BIOL-1

Name: _____ Date: _____

Overview

Gregor Mendel discovered that traits are inherited in predictable patterns. By crossing pea plants and counting thousands of offspring, he uncovered the fundamental rules of genetics — rules that apply to all living things, including humans. In this lab you will master those rules step by step: learn the core vocabulary, practice with Punnett squares, and then apply what you know to increasingly complex inheritance patterns — from simple dominance, to codominance and multiple alleles, to sex-linked and polygenic traits.

Learning Objectives

By the end of this lab, you will be able to:

1. Define essential genetics terms and explain Mendel's two laws in the context of meiosis.
2. Use Punnett squares to predict genotypic and phenotypic ratios for monohybrid and dihybrid crosses.
3. Model allele segregation with a hands-on probability activity.
4. Solve problems involving incomplete dominance and codominance.
5. Determine blood type inheritance using the ABO multiple-allele system.
6. Predict outcomes of X-linked trait crosses and explain why sex-linked traits are more common in males.
7. Draw and analyze a pedigree to determine genotypes and patterns of inheritance.
8. Explain polygenic traits and how environment interacts with genotype to produce phenotype.

Materials

- This worksheet
- Scratch paper for additional Punnett squares

- Two coins (for Part 2)

Part 1: Key Concepts & Mendel's Laws

Before solving genetics problems, you need the language. Study the reference definitions below, then answer the questions that follow.

Reference Definitions

Term	Definition
Gene	A segment of DNA that codes for a specific trait
Allele	A version of a gene (e.g., a brown-eye allele vs. a blue-eye allele)
Diploid	Having two copies of each chromosome — one from each parent — so you carry two alleles for every gene
Genotype	The alleles an organism carries, written with letters (e.g., <i>Aa</i>)
Phenotype	The physical trait you can observe (e.g., "brown eyes")
Homozygous	Two identical alleles (<i>AA</i> or <i>aa</i>)
Heterozygous	Two different alleles (<i>Aa</i>) — also called a carrier if the trait is recessive
Dominant	The allele expressed when at least one copy is present (capital letter, e.g., A)
Recessive	The allele expressed only when two copies are present (lowercase letter, e.g., a)
Punnett Square	A grid used to predict the offspring of a genetic cross

Check Your Understanding

1. The genotype *Aa* contains two letters — one capital, one lowercase — each representing a different allele of the same gene. Using the definitions above, explain in your own words: What does the capital **A** represent? What does the lowercase **a** represent? Why are there exactly two letters (connect this to the concept of diploid)?

2. An organism with genotype *aa* shows the recessive trait. What genotype(s) would show the dominant trait?

3. Why is a person with genotype **Aa** called a "carrier" for a recessive condition?

4. State Mendel's Law of Segregation in your own words. How does it connect to what happens during Meiosis I?

5. State Mendel's Law of Independent Assortment in your own words. What does it assume about the genes being considered?

Part 2: Coin Toss — Modeling Allele Segregation

Learning Goal: Experience how random chance during meiosis determines which allele each parent passes on.

When a parent with genotype **Aa** makes gametes (sperm or eggs), each gamete randomly receives either **A** or **a**. This is like flipping a coin — a 50/50 chance for each allele. By simulating this process, you will see how probability governs inheritance.

Procedure

1. Get **two coins**. One coin represents the **Father (Aa)**. The other represents the **Mother (Aa)**.
2. Flip both coins simultaneously **20 times**. For each flip, record the offspring genotype:
3. **Heads** = Dominant allele (**A**)
4. **Tails** = Recessive allele (**a**)
5. Both Heads = **AA** · One Head + One Tail = **Aa** · Both Tails = **aa**

Data

Genotype Tally (20 Tosses)

#	Genotype	Tally	Count	Percentage (Count ÷ 20 × 100)
1				
2				
3				

Questions

6. The expected ratio for $Aa \times Aa$ is 25% AA : 50% Aa : 25% aa. How close were your results to the expected ratio?

7. If you flipped 100 times instead of 20, would your results be closer to or farther from the expected ratio? Explain why using the concept of sample size.

Part 3: Monohybrid Crosses

Now apply the Punnett square method to predict offspring outcomes for a single trait.

Albinism is a recessive condition in which the body produces little or no melanin pigment.

- **A** = Normal pigmentation (dominant)
- **a** = Albinism (recessive)

Cross 1: Heterozygous $\times$ Homozygous Recessive ($Aa \times aa$)

8. A man with normal pigmentation who is a carrier (Aa) has children with a woman who has albinism (aa). Fill in the Punnett square:

	a	a
A		
a		

• Genotypic ratio:

• Phenotypic ratio:

• What percentage of offspring could have albinism?

9. Can any child from this cross be homozygous dominant (AA)? Why or why not?

Cross 2: Heterozygous $\times$ Heterozygous ($Aa \times Aa$)

10. Two carriers ($Aa \times Aa$) have children. Fill in the Punnett square:

	A	a
A		
a		

• Genotypic ratio:

• Phenotypic ratio:

• What fraction of offspring could have albinism?

- What fraction would be carriers (Aa) but appear to have normal pigmentation?

11. What is a test cross, and what is it used for? If you crossed an organism showing the dominant phenotype with a homozygous recessive organism, what phenotypic ratio would you expect if the dominant-phenotype parent is heterozygous?

Part 4: Dihybrid Crosses

When two traits are tracked at once, we use a 4×4 Punnett square. Mendel's Law of Independent Assortment tells us the two traits are inherited independently (provided the genes are on different chromosomes).

In pea plants:

- **Tall (T)** is dominant over **short (t)**
- **Round seed (R)** is dominant over **wrinkled seed (r)**

12. Cross two plants that are both heterozygous for both traits ($TtRr \times TtRr$). What is the expected phenotypic ratio? (Hint: Consider the four possible phenotypic categories.)

13. Out of 16 offspring expected in this cross, how many would you expect to be Tall with Round seeds?

14. What law of Mendel does the dihybrid cross demonstrate? How does this law connect to what happens during Metaphase I of meiosis when homologous chromosomes line up at the cell plate?

Part 5: Incomplete Dominance

In incomplete dominance, neither allele is fully dominant — the heterozygote shows a *BLEND* of both traits.

Example: In snapdragons, Red flowers (RR) crossed with White flowers (R'R') produce **Pink** flowers (RR').

15. Cross two pink snapdragons (RR' × RR'). Fill in the Punnett square:

	R	R'
R		
R'		

• Phenotypic ratio: Red :
 Pink :
 White

16. How is incomplete dominance different from complete dominance? In complete dominance, what would the heterozygote look like?

17. How is incomplete dominance different from codominance? (Hint: In codominance, *both* alleles are fully expressed at the same time rather than blending.)

Part 6: ABO Blood Types — Codominance & Multiple Alleles

The ABO blood system is a real-world example of both codominance (I^A and I^B are codominant — both expressed) and multiple alleles (three alleles exist in the population: I^A , I^B , i). This system determines which antigens appear on your red blood cells and is critical for safe blood transfusions.

Blood Type Reference Table

Blood Type	Possible Genotype(s)	Antigens on Red Blood Cells	Antibodies in Plasma
Type A	$I^A I^A$ or $I^A i$	A antigen	Anti-B
Type B	$I^B I^B$ or $I^B i$	B antigen	Anti-A
Type AB	$I^A I^B$	Both A and B	Neither
Type O	ii	Neither	Anti-A and Anti-B

18. Using the reference table above, explain why Type AB blood is an example of codominance.

19. A mother is Type A ($I^A i$) and a father is Type B ($I^B i$). Fill in the Punnett square:

	I^B	i
I^A		
i		

20. What blood types are possible in their children? List all possible blood types.

21. Can they have a child with Type O blood? Explain using the Punnett square above.

22. Why is a person with Type O blood sometimes called a "universal donor"? (Hint: Think about what antigens are on Type O red blood cells.)

Part 7: Sex-Linked Traits

Some genes are located on the X chromosome. Because males have only one X (XY), a single recessive allele on the X is enough to cause the trait. Females (XX) need two copies of the recessive allele.

Color blindness (red-green type) is a recessive X-linked trait.

Sex-Linked Genotype Reference

Genotype	Sex	Phenotype
$X^B X^B$	Female	Normal vision
$X^B X^b$	Female	Normal vision (carrier)
$X^b X^b$	Female	Color blind
$X^B Y$	Male	Normal vision
$X^b Y$	Male	Color blind

Cross: Carrier Mother $\times$ Normal Father

A mother with normal vision who is a **carrier** ($X^B X^b$) has children with a father who has **normal vision** ($X^B Y$).

23. Fill in the Punnett square:

	X^B	Y
X^B		
X^b		

24. What is the probability of having a color-blind son?

25. What is the probability of having a color-blind daughter?

26. What fraction of their daughters would be carriers?

27. In your own words, explain why X-linked recessive traits (like color blindness and hemophilia) are much more common in males than in females.

Part 8: Pedigree Analysis — Tracing a Trait Through Generations

Learning Goal: Read and draw a pedigree chart to trace how a genetic trait passes through a real family scenario.

A **pedigree** is a diagram that shows how a trait passes through a family across generations. Pedigrees are a key tool in genetic counseling.

Standard Pedigree Symbols

Symbol	Meaning
□	Male
○	Female
■ / ●	Affected (shows the trait)
Half-shaded □ or ○	Carrier (has one copy of recessive allele)
□—○	Mated pair
Vertical line down	Their children

Scenario: Cystic Fibrosis

Cystic fibrosis (CF) is an **autosomal recessive** condition (genotype **ff**).

- **F** = Normal lung function (dominant)
- **f** = Cystic fibrosis (recessive)

Read the family description, then **draw the pedigree** in the box below.

Generation I (Grandparents): Frank (unaffected) and Gina (unaffected) have three children.

Generation II (Their children):

- **Maria** (unaffected) — marries **Tom** (unaffected). They have 2 children.
- **Carlos** — has cystic fibrosis.
- **Anna** (unaffected) — no children yet.

Generation III (Grandchildren):

- Maria and Tom's daughter **Lily** (unaffected)
- Maria and Tom's son **Sam** (unaffected)

Draw the Pedigree

Use the standard symbols above. Squares for males, circles for females. Shade in anyone who is affected. Label each person with their name.



Pedigree Analysis

28. Carlos has cystic fibrosis (ff). He received one *f* allele from Frank and one from Gina. What must Frank and Gina's genotypes be?

29. Frank and Gina appear healthy but each carry one recessive allele. What is the genetic term for this?

30. What are Maria's possible genotypes? (There are two possibilities — list both.)

31. If Maria is a carrier (Ff) and Tom is also a carrier (Ff), what is the probability that their child would have cystic fibrosis? Draw the Punnett square below to support your answer.

Probability:

32. Looking at the pedigree, could you tell just by looking at appearances that Frank and Gina were carriers? Why is this important for genetic counseling?

Part 9: Polygenic Traits & the Environment

Not all traits follow simple Mendelian patterns. Many human traits — like height, skin color, and eye color — are controlled by many genes working together. These are called polygenic traits.

33. What is a polygenic trait? Name two examples in humans.

34. Mendel's pea plants were either tall OR short — two distinct categories. Human height, however, shows a continuous range from very short to very tall and follows a bell curve. Why does human height show a continuous distribution instead of two discrete categories? (Hint: Think about the number of genes involved and how their effects add up.)

35. True or False: If you know someone's genotype, you can always predict their exact phenotype. Explain your answer, giving a specific example where an environmental factor changes the phenotype.

36. A student says: "Genes determine everything about you — environment doesn't matter." Do you agree or disagree? Support your answer with a specific example showing how genes and environment interact to produce a phenotype.

Conclusion

37. In your own words, explain why two healthy parents can have a child with a recessive genetic condition like albinism, cystic fibrosis, or sickle cell disease.

38. This lab covered many different inheritance patterns: simple dominance, incomplete dominance, codominance, sex-linked inheritance, and polygenic traits. Pick one pattern and explain in 2–3 sentences how it differs from simple Mendelian dominance.

39. What is one thing you learned in this lab that changed or deepened your understanding of how traits are inherited?